



UNIVERSITÀ DI PISA

Oceanus Fleet Watch

Visual Analysis of Suspicious Activities in the Oceanus Fleet

Version 2 — Redesign

VAST Challenge 2024 Data Analysis - Mini-Challenge 2

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Companion to the original report (Version 1, November 7, 2025)

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1 Introduction: Why a Second Version

1.1 Context

The VAST Challenge 2024 Mini-Challenge 2 asks analysts to help the non-profit **FishEye International** investigate illegal fishing in the fictional Oceanus archipelago, with **SouthSeafood Express Corp** as the main suspect. Version 1 of this project (*reports/Report_VAST2024_v1.pdf*, code in *v1-original/*) answered the challenge with a Leaflet map, a force-directed graph and a timeline. This report documents **Version 2**, called *Oceanus Fleet Watch* (code in *v2-redesign/*): a redesign of the same dashboard that also re-examines the analysis underneath it.

The redesign started as a visual update. While rebuilding the views, however, the data pipeline was re-checked ping by ping, and two defects in how v1 computed its *suspicious signature* came to light (Section 2.2). Fixing them changed the answers to the challenge questions, so Version 2 is as much an analytical correction as a visual one. Both versions are kept side by side in the repository so the difference can be inspected directly.

1.2 Related work

As discussed in the first report, work on illegal, unreported and unregulated (IUU) fishing splits into automated detection, such as Global Fishing Watch’s AIS and satellite analytics [4, 5] and machine-learning behaviour classifiers [6], and human-in-the-loop visual analytics that combine maps, networks and timelines [7, 8]. Both versions of this dashboard belong to the second family [3]. Version 2 adds two ideas from outside that literature: the visual conventions of official nautical charts [13] and an explicit, testable data pipeline.

1.3 What changed at a glance

Aspect	Version 1	Version 2
Runs from the repository	No: needs the 138 MB <i>mc2.json</i> (only inside the zip) plus two images that were never committed	Yes: a 3.5 MB derived file ships in <i>v2-redesign/data/</i>
Pings used by the analysis	106,229 of 258,542 (41%)	258,542 of 258,542 (100%)
“In a preserve” test	Buoy points tested with <i>d3.geoContains</i> ; wrong polygon winding makes every buoy match	Ping location read directly (the graph already names the preserve)
Suspicion metric	Count of buoy pings	Hours of dwell time inside the three ecological preserves
Map	Leaflet + OpenStreetMap tiles (empty ocean under a fictional archipelago)	D3 nautical chart drawn from the GeoJSON, with relief, soundings, scale bar
Network view	Force graph where every company links to one “Forbidden zone” node	Company → preserve flow diagram with the three real preserves
Detail on demand	Text panel	Ranked suspects → company → vessel → visits; preserve pages with species
Themes / devices	Light only, desktop only	Light and dark, desktop and mobile

Table 1: Summary of the differences between the two versions.

2 Data and Analysis Corrections

2.1 From a knowledge graph to an analysis file

Version 1 loaded the full knowledge graph in the browser and, for every ping, searched the node and geography lists with *Array.find*. That approach has two costs: the page cannot be served from GitHub (the file exceeds the 100 MB limit) and every filter change repeats hundreds of thousands of linear searches. Version 2 moves the expensive work offline. A small script, `v2-redesign/scripts/build_data.py`, reads `mc2.json` once and writes `oceanus.json`: vessels, locations with coordinates (polygon centroids for regions), and all pings as sorted columnar arrays (location index, minutes since 1 Jan 2035, dwell minutes). The browser then makes a single pass over the 258,542 pings per date range and every view reads from those totals.

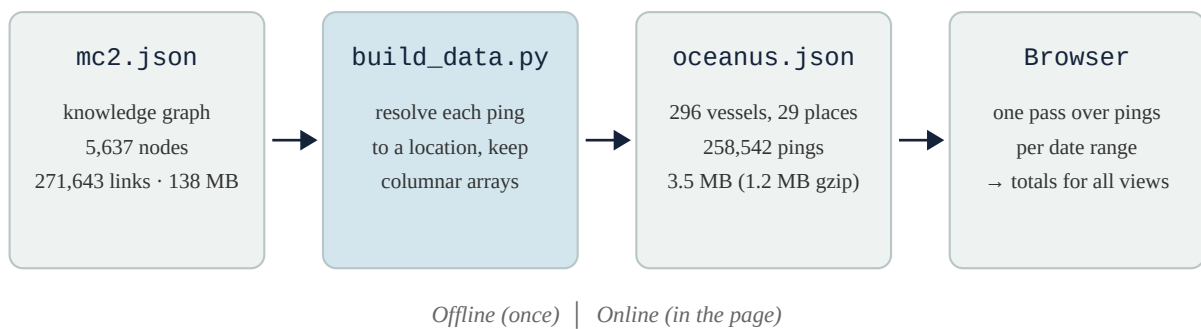


Figure 1: Data pipeline of Version 2.

2.2 Two defects in the Version 1 signature

To compare the versions fairly, the exact v1 logic was re-run on the raw data (same files, same *d3.geoContains* call). The result is summarised in Table 2.

Step in the v1 pipeline	Pings	Share
Transponder pings in <i>mc2.json</i>	258,542	100%
Dropped: source is “City of X”, but the nodes file uses “X”	24,818	9.6%
Dropped: source is a region (e.g. <i>Nemo Reef</i>), but only Point features are looked up	127,495	49.3%
Resolved to a buoy position	106,229	41.1%
Of those, flagged as “inside a preserve”	106,229	100% of resolved

Table 2: Re-running the Version 1 suspicion logic on the full dataset.

Defect 1 – lost pings. The pings that matter most were the ones discarded: a ping whose source is *Nemo Reef* or *Ghoti Preserve* is a ping inside a preserve, but v1 only accepted locations that resolve to a GeoJSON Point, so every region ping (49% of the data) was dropped before any test.

Defect 2 – every buoy was “inside” a preserve. *d3.geoContains* works on the sphere and uses ring winding to decide which side of a polygon is the inside. The Oceanus GeoJSON follows RFC 7946 (counter-clockwise exterior rings) [9], which D3 reads as “the whole globe except this small area”

[10]. Every buoy therefore tested positive, including *Exit East* at the edge of the map. The v1 ranking was, in practice, a ranking of buoy traffic: its top suspect *Ritter Ltd* (3,933 flagged pings) ranks 27th of 83 once real preserve time is measured (644 h).

2.3 The revised signature

Version 2 keeps the two indicators of v1 but measures them on all the data:

- **Preserve dwell time.** For each fishing vessel, the dwell minutes of every ping located in one of the three ecological preserves (Ghoti Preserve, Nemo Reef, Don Limpet Preserve). The GeoJSON lists only recreation, tourism and research as activities in these areas, so commercial dwell there is a meaningful red flag. Companies are ranked by total hours.
- **Transponder gaps.** More than 12 hours between the *end* of one ping (time + dwell) and the next ping. Version 1 measured the time between ping start times, which flagged ordinary long stays in port as gaps.

Across the whole period the 179 company-owned vessels log 18,837 pings and 48,470 hours inside preserves; 83 of 99 companies appear at least once. Nemo Reef alone accounts for 33,944 hours (70%), Ghoti Preserve for 14,519 and Don Limpet Preserve for only 6.

3 Design Choices

The Shneiderman mantra that guided v1 – overview first, zoom and filter, then details on demand [1] – is kept, but every element was redesigned around one subject: *a nautical chart on an investigator's desk*. The visual work followed two open design guides adopted as coding “skills”: Emil Kowalski's design-engineering principles [11] and the *taste-skill* redesign audit [12].

3.1 Layout and interface

The left control column of v1 is gone. Filters live in a compact toolbar in the header (search for a company *or* a vessel, top-N switch, date range, reset), freeing the width for data. Below it, a KPI strip summarises the current scope. The board below uses a grid: the chart of Oceanus as the main view, the suspect panel on the right, and the timeline and flow diagram underneath (Figure 2). On phones the same content stacks into one scrolling column.

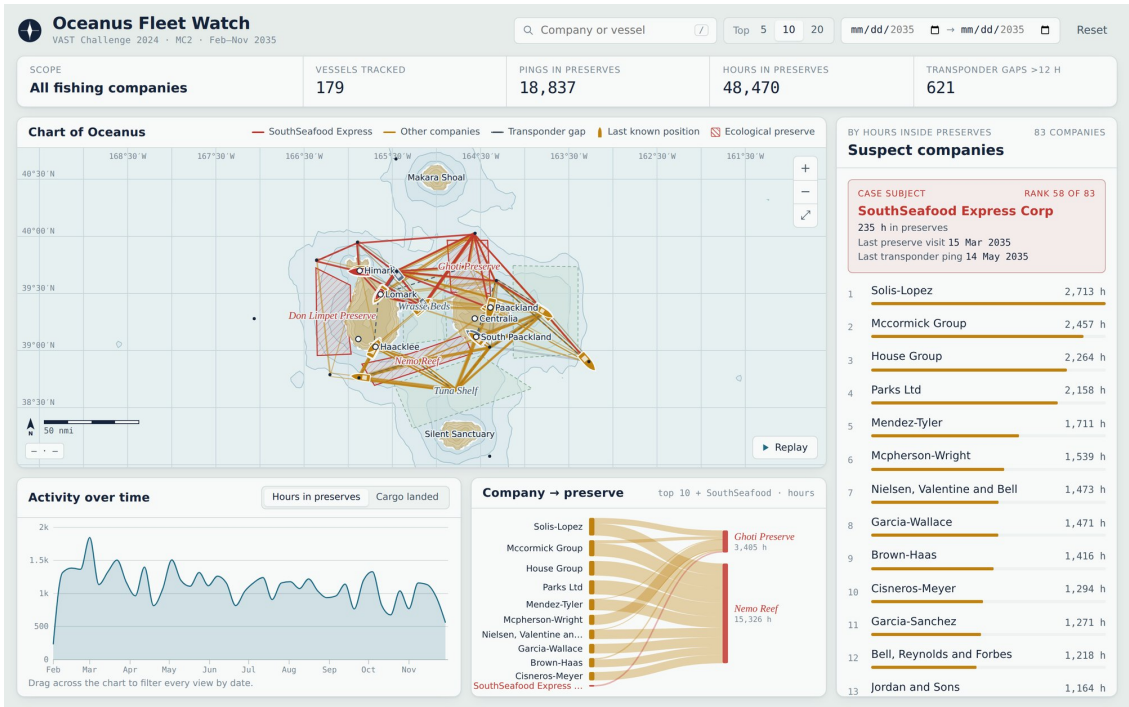


Figure 2: Version 2 overview (light theme). Toolbar and KPI strip on top, chart of Oceanus, suspect ranking with the pinned case card, activity timeline and company → preserve flows.

3.2 Colors

Color is managed as a small set of design tokens defined once as CSS custom properties, with a light “chart paper” palette and a dark “night bridge” palette (Figure 3). The semantic roles of v1 are kept but softened: **red** is reserved for SouthSeafood Express, **amber** for every other company with preserve time, a muted slate for vessels without it, and a single interactive accent (**sea blue**) for controls and selections. The saturated v1 blue gradient and the tiled fish wallpaper were removed: they competed with the data.



Figure 3: Core design tokens (left half light, right half dark). The same tokens exist as variables in the Figma file (Section 7).

Following nautical-chart convention [13], *shallow* water is the darker blue and deep water the lighter tone, the opposite of a generic “darker = deeper” ramp. Preserves use a red hatch rather than a solid fill so tracks stay visible through them.

3.3 Typography

Three families from the IBM Plex superfamily plus one serif, each with a job: **IBM Plex Sans Condensed** for titles, **IBM Plex Sans** for interface text, **IBM Plex Mono** with tabular figures for every number (KPIs, hours, coordinates), and **Newsreader Italic** for water features on the chart, the hydrographic convention for sea areas. The uppercase, text-shadowed Titillium title of v1 was replaced by a quieter title in sentence case.

3.4 Nautical chart realism

Instead of photographs, which would hide the data and add megabytes, realism comes from cartographic detail (Figure 4). The island outlines are rasterised to an offscreen canvas, a two-pass chamfer distance transform gives the distance of every cell to the coast, and that distance, modulated by fractal value noise, becomes a depth field at sea and an elevation field on land. D3 contouring turns both fields into depth bands with isobaths at 5–150 m and island terrain with contour lines. Fishing grounds and preserves are made shallower (shoals and reefs). A white surf line follows the coast, spot soundings appear when zooming in, and a north arrow and a nautical-mile scale bar follow the zoom level. Vessels are drawn as top-down hull silhouettes, rotated to their last heading and scaled by their registered length. The seabed is generated for context and is labelled as such; it is not survey data.

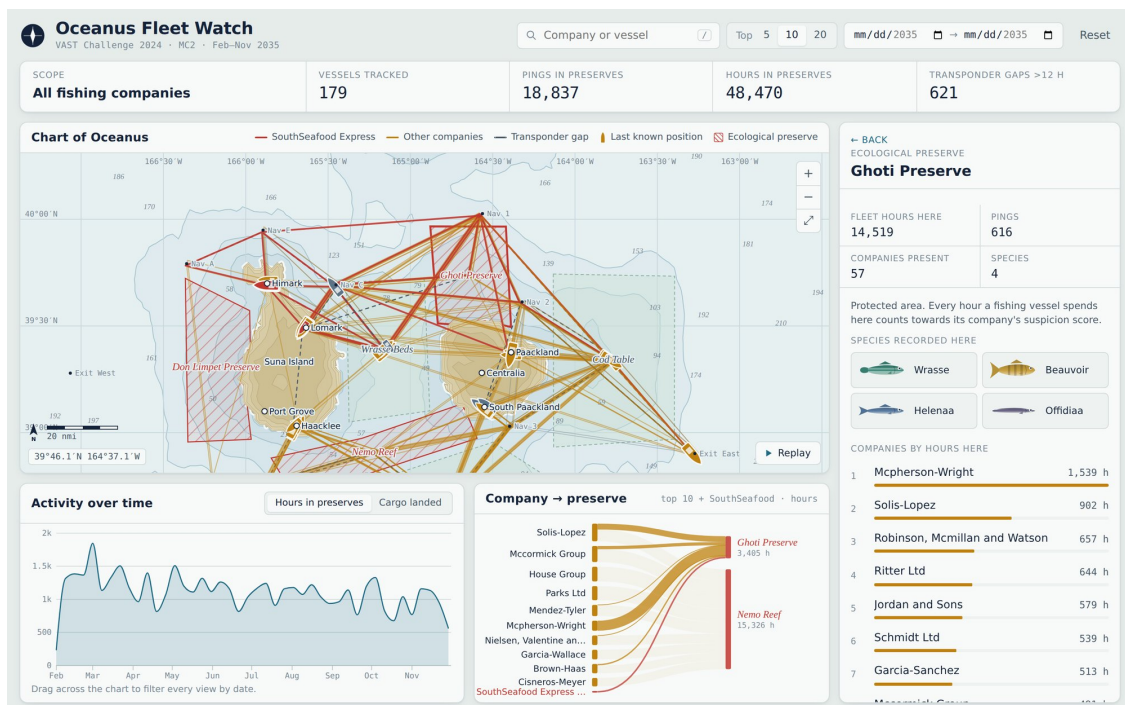


Figure 4: Zoomed chart with bathymetry, isobaths, soundings and island terrain; the side panel shows the Ghoti Preserve page with illustrated species and the companies that spent time there.

3.5 Motion and interaction polish

Motion is used sparingly and always has a purpose [11]. Panel changes fade in over 180 ms with a strong ease-out curve; buttons scale to 97% on press; hover effects are limited to devices with a fine pointer; keyboard actions (/ to search, *Esc* to go back) are never animated; and *prefers-reduced-motion* removes positional movement. The replay (Section 4.6) is the one deliberate, linear animation, because its job is to show time passing. Loading shows skeleton placeholders, empty selections show a short explanation, and a failed load explains how to serve the files.

4 Detailed Visualization Description and Interaction

4.1 KPI strip

Five figures for the current scope (all companies, one company or one vessel) and date range: vessels tracked, pings in preserves, hours in preserves and transponder gaps. The scope name turns red when SouthSeafood is selected.

4.2 Chart of Oceanus

- **Visualization:** islands, fishing grounds (dashed green), hatched preserves, port cities and buoys; tracks for the vessels in scope, colored by the semantic palette; dashed lines for transponder gaps; boat markers at last known positions; graticule labelled in degrees and minutes and a live cursor readout.
- **Interaction:** scroll or buttons to zoom, drag to pan; hovering a track or boat highlights it and dims the rest; clicking it opens that vessel; clicking a preserve or fishing ground opens its page. Labels are placed greedily by priority so they never overlap.

4.3 Suspect ranking and drill-down

The right panel replaces v1's text box with a navigable hierarchy. The default view ranks companies by hours inside preserves with inline bars, and pins a *case card* for SouthSeafood that shows its rank, preserve hours, last preserve visit and last transponder ping whatever its position in the list. A company page lists its vessels and hours per preserve; a vessel page lists its preserve visits with dates and durations (Figure 5); a preserve page shows the recorded species as illustrations and the companies present. Every level has a back link, mirrored by *Esc*.

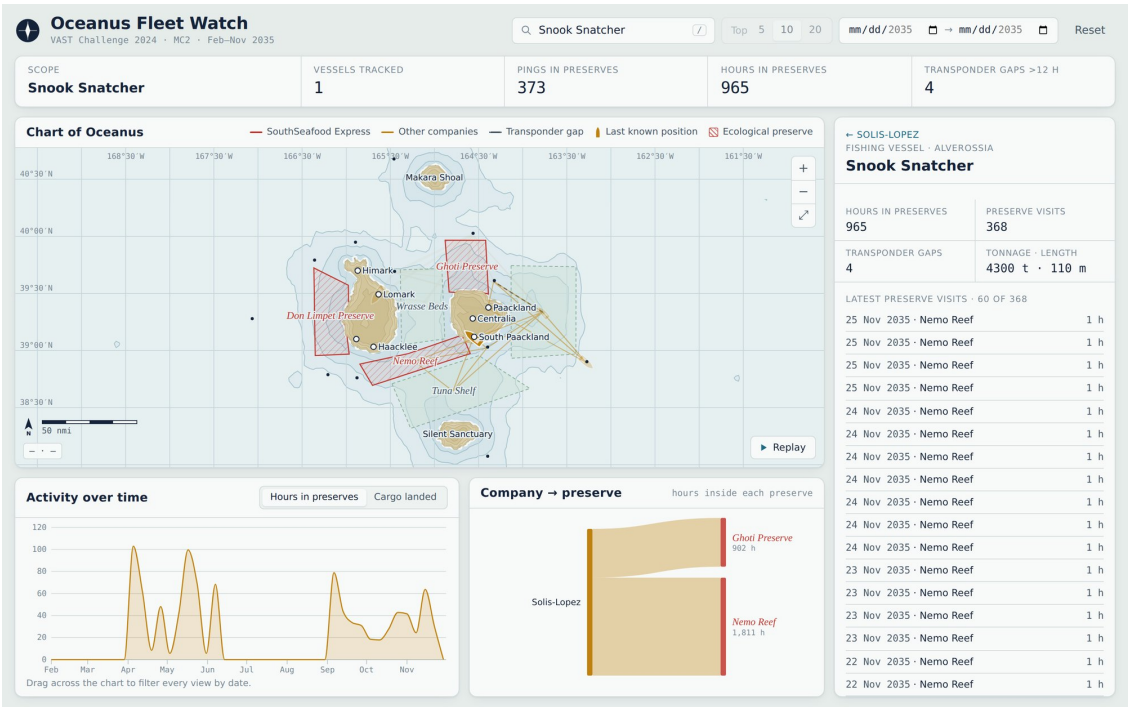


Figure 5: Vessel page for Snook Snatcher (Solis-Lopez, the top-ranked company): 965 h in preserves over 368 visits. The timeline shows two fishing seasons, April–June and September–November.

4.4 Activity over time

A weekly area chart of hours inside preserves for the current scope, switchable to cargo landed (tons). A visible brush selects a date range that filters every view, and the date fields in the toolbar stay in sync with it (Figure 6). When a company is selected in cargo mode, a note explains that cargo records are not linked to vessels.

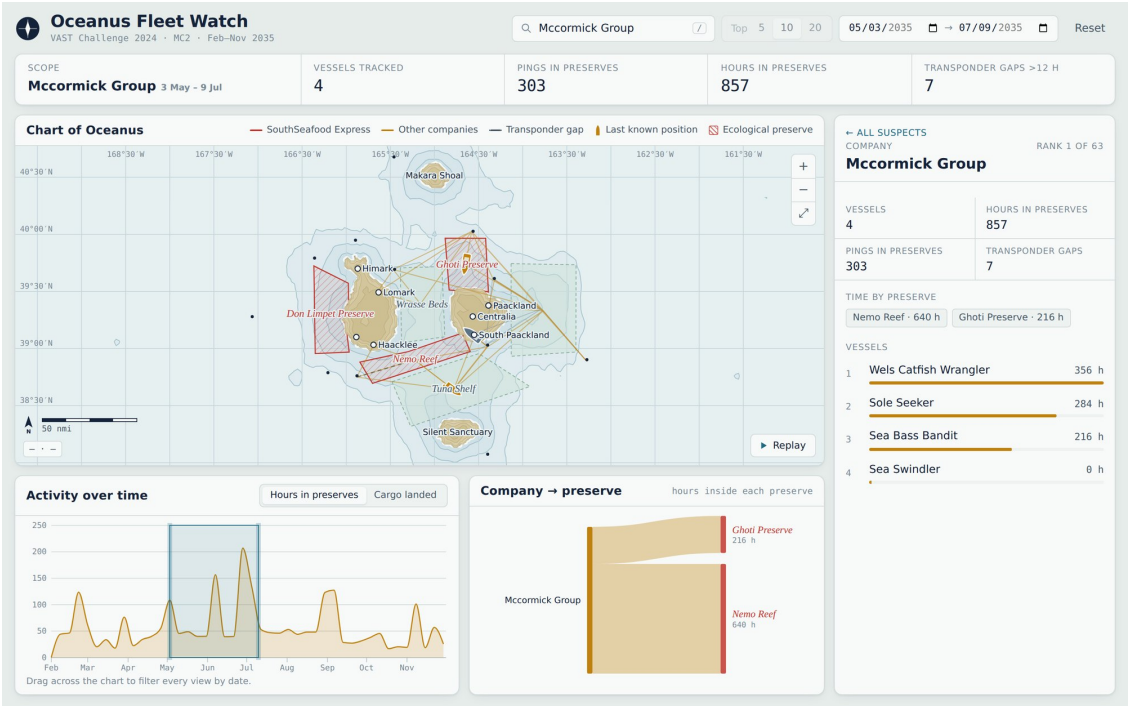


Figure 6: McCormick Group filtered to a brushed period; map, KPIs, ranking and flows all respond.

4.5 Company → preserve flows

In v1 every company node linked to a single “Forbidden zone” node, so the force layout was a star that carried no extra information. Version 2 uses a bipartite flow diagram: companies in scope on the left, the real preserves on the right, band width proportional to hours. It shows at a glance that most preserve time is spent in Nemo Reef and which companies prefer Ghoti Preserve. Hovering isolates a flow; clicking opens the company or the preserve.

4.6 Replay

The Replay button sails every vessel in scope along its pings through the selected period in about 16 seconds, interpolating between locations and holding position for the recorded dwell time, with the simulated date and time shown next to the button (Figure 7). Tracks are dimmed during playback so the boats read clearly.

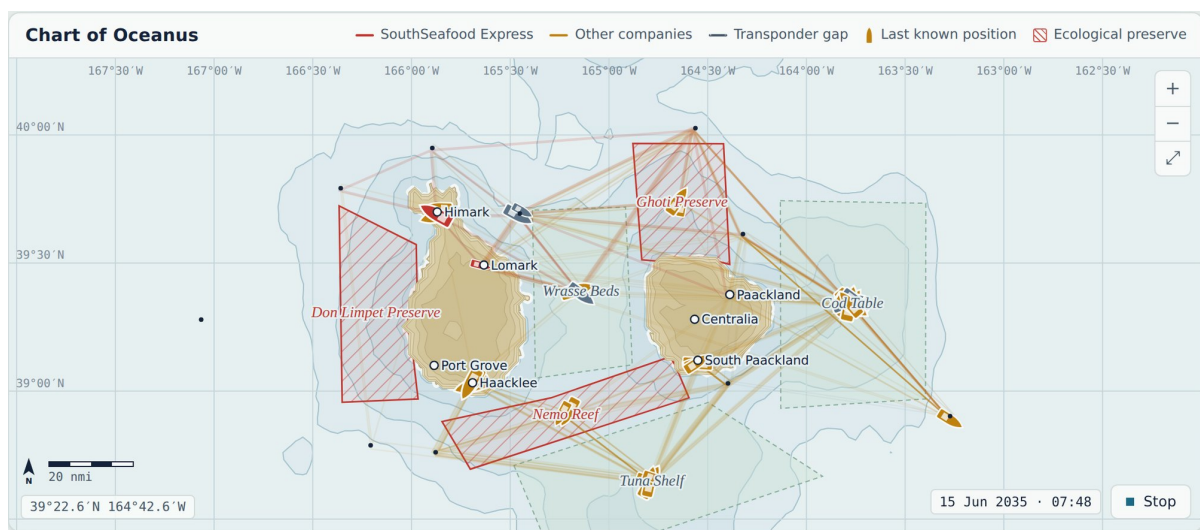


Figure 7: Replay in progress on a zoomed chart; boats are oriented along their course.

4.7 Coordinated analytical workflow

1. **Overview:** the KPI strip, ranking and flows show where preserve time goes; the case card keeps SouthSeafood in view.
2. **Filter:** pick a company from the ranking, the flows, the map or the search box; every view switches scope.
3. **Zoom in time:** brush the timeline or type dates; all totals are recomputed for the period.
4. **Details on demand:** open a vessel for its visits, or a preserve for species and companies; replay to see movement.

5 Use Case Examples for the Analytical Tasks

5.1 Task 1 (Q1/Q3): the signature of SouthSeafood Express Corp

1. **Overview:** the case card shows at once that SouthSeafood ranks only 58th of 83 by preserve hours (235 h). Version 1 suggested the opposite impression because every buoy ping counted as suspicious.

2. **Filter:** clicking the card (Figure 8) shows two vessels. *Snapper Snatcher* made 4 visits to Ghoti Preserve totalling 235 h between 2 February and 15 March 2035; *Roach Robber* never entered a preserve.
3. **Time:** the timeline drops to zero after mid-March. The case card and vessel pages show that both vessels stop transmitting altogether on 12–14 May 2035, and the company logs 3 transponder gaps.
4. **Conclusion:** SouthSeafood's signature is not volume but pattern: concentrated early visits to Ghoti Preserve, the preserve with Wrasse, Beauvoir, Helena and Offidia, followed by complete silence. This is consistent with the challenge narrative of a company that was caught and ceased operating, and it explains why a volume ranking alone would miss it.

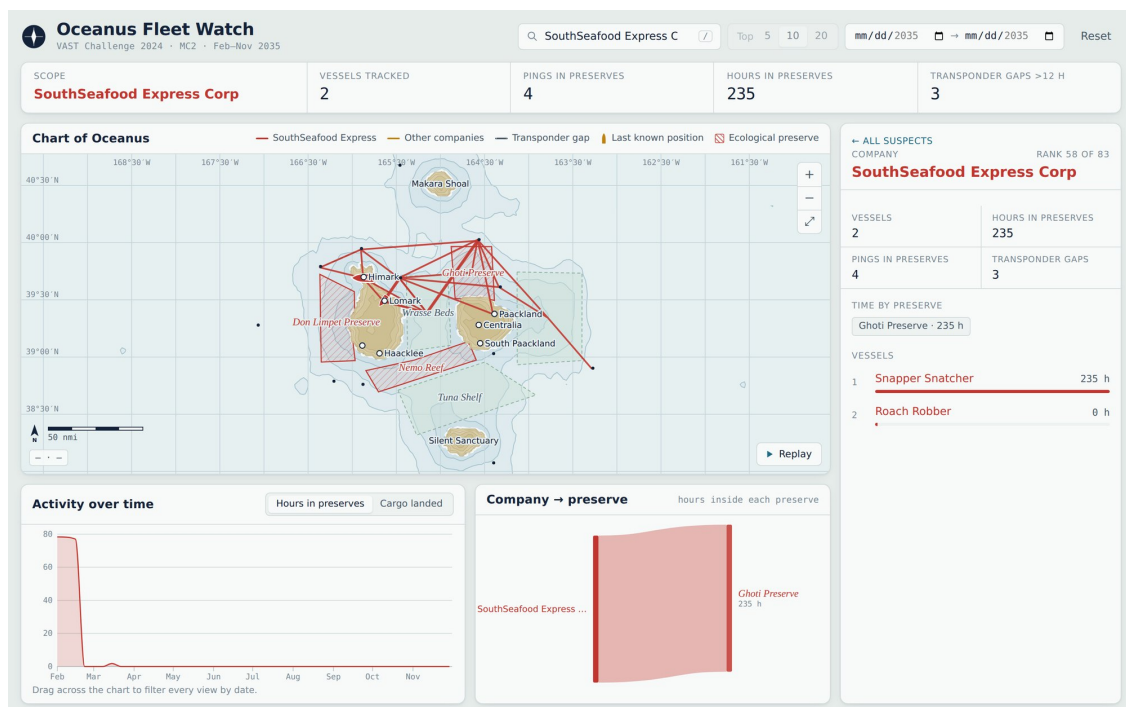


Figure 8: Dashboard filtered for SouthSeafood Express Corp. All preserve time falls in February and early March and is spent in Ghoti Preserve.

5.2 Task 2 (Q4): other actors with similar behaviour

1. **Reset and rank:** the leading companies are Solis-Lopez (2,713 h), McCormick Group (2,457 h), House Group (2,264 h), Parks Ltd (2,158 h) and Mendez-Tyler (1,711 h); the top ten account for 38% of all preserve time.
2. **Compare patterns:** companies whose flows lean towards Ghoti Preserve, SouthSeafood's preserve, are the closest behavioural matches; opening Ghoti Preserve lists them by hours (Mcpherson-Wright 1,539 h, Solis-Lopez 902 h).
3. **Drill down:** Solis-Lopez's vessel *Snook Snatcher* (Figure 5) shows repeated short visits in two seasons, a regular pattern very different from SouthSeafood's burst-then-silence.
4. **Conclusion:** the analyst obtains a ranked, reproducible shortlist for FishEye and can explain each entry with dates, places and hours instead of a single aggregated count.

6 Before and After

Figure 9 places the two versions side by side at the same task. The v1 screenshot is taken from the original report.

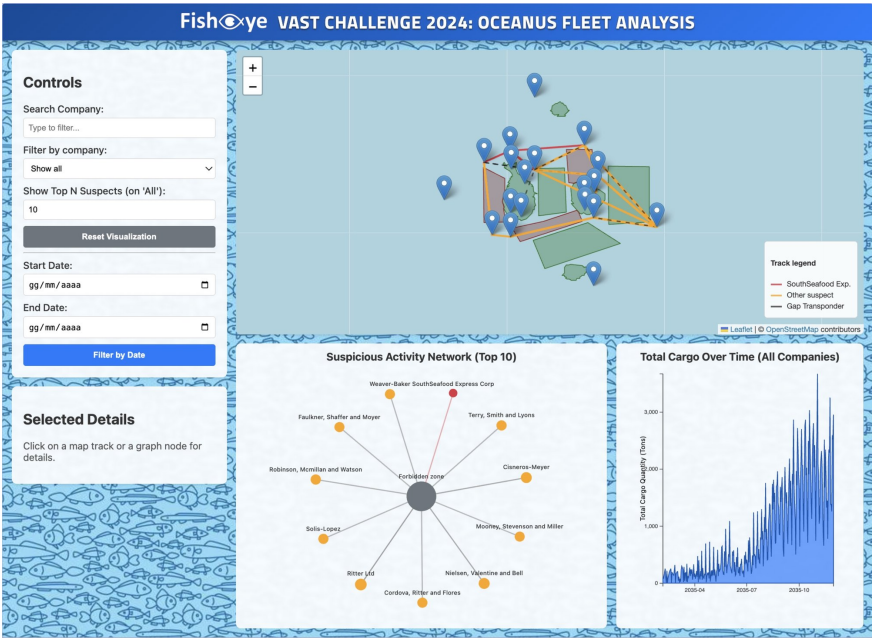


Figure 9a: Version 1 – Leaflet map on OpenStreetMap tiles, force graph around a single “Forbidden zone” node, total-cargo timeline.

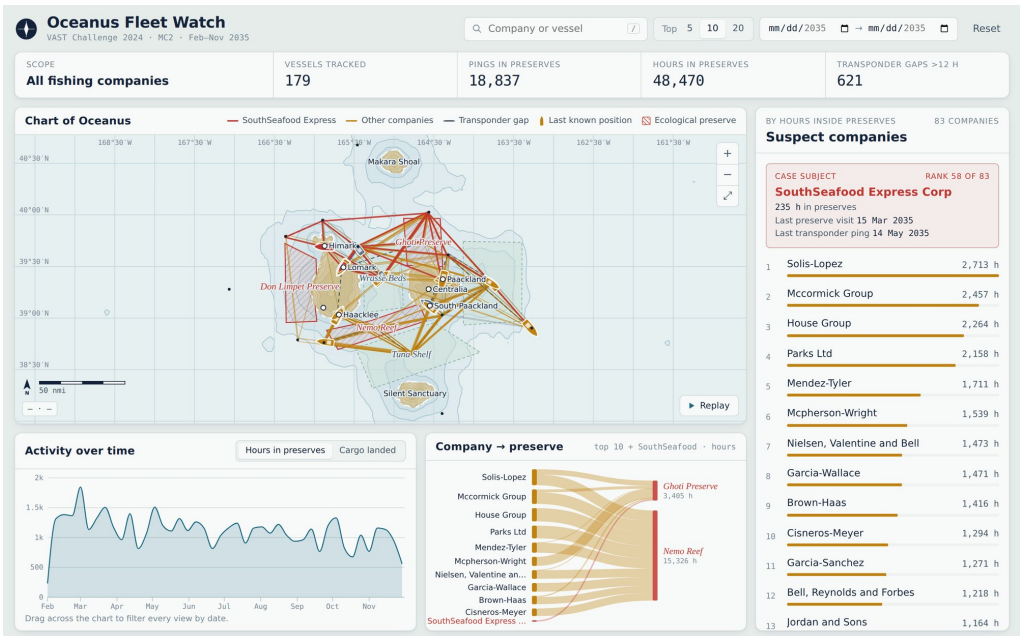


Figure 9b: Version 2 – the same questions answered with corrected data and a redesigned interface.

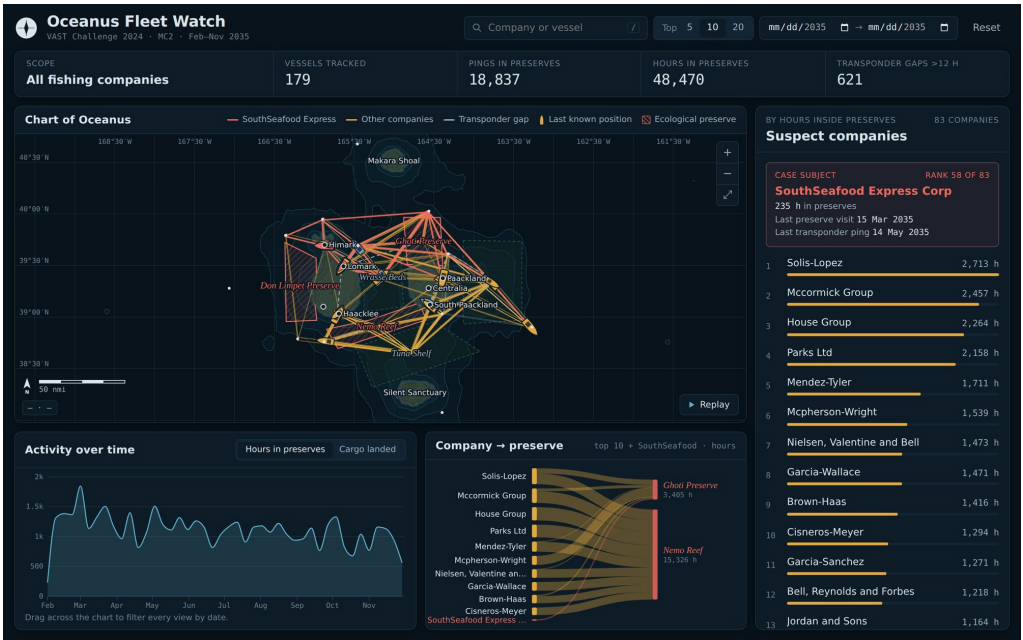


Figure 10: Version 2 in the dark theme. The relief colors are recomputed from the dark tokens.

Version 2 also works at phone width (Figure 11): the toolbar wraps, KPIs form a two-column grid, the chart keeps its controls and the panels stack in reading order.

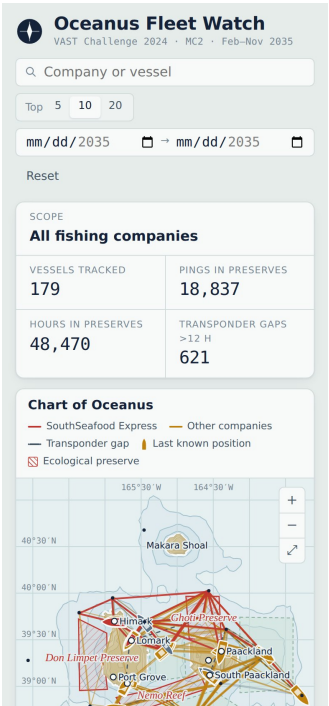


Figure 11: Version 2 on a 390 px wide phone screen.

7 Design System in Figma

The website does not need Figma to run: the code is the product. Figma plays a supporting role as the design record of Version 2. The file *Oceanus Fleet Watch — Redesign* contains two variable collections (*Oceanus Light* and *Oceanus Dark*, the same tokens as Figure 3), eleven text styles matching the CSS type scale, a token sheet, and the full dashboard rebuilt with native auto-layout frames in both themes, with the charts imported as editable vectors. Its uses are practical:

- **Collaboration and review:** a supervisor or designer can comment on the layout without reading code.
- **Single source of truth for style:** tokens and text styles can be changed in one place before touching CSS.
- **Communication:** clean frames for slides and reports, and a starting point for future iterations.

The Figma file reflects the first v2 layout. The later cartographic layer (relief, boats, species) was designed directly in code, where it is generated from data.

8 Implementation Notes

Path	Content
v1-original/	Version 1 exactly as submitted (HTML, CSS, JS) and a helper that extracts its raw data from the zip
v2-redesign/index.html	Page structure: toolbar, KPI strip, four panels, map controls
v2-redesign/style.css	Tokens for light and dark themes, layout grid, components, responsive rules
v2-redesign/script.js	Data model, aggregation, map with relief and replay, side panel, timeline, flows
v2-redesign/scripts/build_data.py	Offline conversion of mc2.json into data/oceanus.json
reports/	Report_VAST2024_v1.pdf (original) and Report_VAST2024_v2.pdf (this document)

Table 3: Repository layout after the redesign.

The page uses plain HTML, CSS and JavaScript with D3 v7 [2] as the only library; there is no build step. It was tested in Chromium with Playwright at desktop and phone sizes, in both themes, covering search, drill-down, brushing, zooming, replay and preserve pages, with no console errors. Rendering the relief takes a fraction of a second and happens only when the chart is resized or the theme changes. Interface elements are real buttons with visible focus states, and the layout has no horizontal scrolling at 390 px.

9 Conclusions, Limitations and Reflections

Reflection (Analysis): the most valuable outcome of the redesign was not visual. Re-implementing each view forced every number to be traced back to the raw data, which exposed the dropped pings and the polygon-winding issue. The corrected signature changes both answers: SouthSeafood is identified by its pattern rather than its volume, and the list of similar actors is now driven by real time spent in protected areas.

Reflection (Design): treating the dashboard as a nautical chart gave every decision a reference, from the shallow-is-darker water palette to italic water names, soundings and the scale bar. Replacing the force graph with a flow diagram shows that a simpler, task-specific chart can carry more information than a generic network layout.

Limitations:

- Dwell time inside a preserve is a strong indicator, not proof of fishing: vessels may transit Nemo Reef.
- Ping locations are coarse (29 places); tracks connect place centroids rather than real positions.
- Cargo records are still not linked to vessels, so Q2 is answered only at fleet level, as in v1.
- The seabed and island terrain are procedurally generated for readability and are not measured data.

Reflection (Tools): Version 2 was developed with the help of an AI coding assistant (Claude Code), guided by the design-skill collections cited above. Every analytical figure in this report was verified against the data, and the Version 1 comparison was obtained by re-running the original logic unchanged.

References

- [1] Shneiderman, B. (1996). *The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations*. In Proceedings of the IEEE Symposium on Visual Languages (pp. 336–343).
- [2] Bostock, M., Ogievetsky, V., & Heer, J. (2011). *D3: Data-Driven Documents*. IEEE Transactions on Visualization and Computer Graphics, 17(12), 2301–2309.
- [3] VAST Challenge 2024. *Mini-Challenge 2: Oceanus*. Available online.
- [4] Global Fishing Watch. (2024). *Joint Analytical Cell: Data, Intelligence and Tools to Combat Illegal Fishing*. Available online: <https://globalfishingwatch.org/>.
- [5] Welch, H., Kroodsma, D., & Miller, N. A. (2022). *Hot Spots of Unseen Fishing Vessels*. Science Advances, 8(32), eabm5307.
- [6] Zhao, Q., Chen, Y., & Liu, H. (2024). *Fishing Vessel Behavior Classification Using CNN-LSTM Models*. Applied Sciences, 14(5), 2294.
- [7] Park, A. J., & Stamato, S. Z. (2020). *Social Network Analysis of Global Transshipment: A Framework for Discovering Illegal Fishing Networks*. In IEEE/ACM ASONAM 2020.
- [8] Rodríguez, J. P., Fernández, E., & Pérez, D. (2022). *Identification of Suspicious Behaviour Through Anomalies in the Tracking Data of Fishing Vessels*. arXiv:2204.06781.
- [9] Butler, H., Daly, M., Doyle, A., Gillies, S., Hagen, S., & Schaub, T. (2016). *The GeoJSON Format*. RFC 7946, IETF.
- [10] Bostock, M. *d3-geo: Geographic projections, spherical shapes and spherical trigonometry (polygon winding order)*. <https://d3js.org/d3-geo>.
- [11] Kowalski, E. *Skills for Designers and Engineers*. <https://github.com/emilkowalski/skills>.
- [12] leonxlnx. *taste-skill: redesign and frontend taste skills*. <https://github.com/leonxlnx/taste-skill>.
- [13] NOAA & NGA. *U.S. Chart No. 1: Symbols, Abbreviations and Terms used on Paper and Electronic Navigational Charts*.